

Fractal Information Theory Project

FIN ST233–ST247

Typed Second Operators, Nonlinear Carrier Obstructions,
and Refinement Moduli

Analytical, interval, and computational research report

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Abstract

Programs ST233–ST247 execute the next local analytical and computational research batch after the single-generator obstruction of ST218–ST232. The principal positive result is that a genuine noncommuting second operator can be constructed in a typed way from a vertex state: $\rho \mapsto M_\rho = \text{diag } \rho$. For a localized strict heat state, $[A, M_\rho] \neq 0$, and layer order becomes observable. The equally important negative result is that localization and orientation are then carried by the conditioning state; they are not generated by the fully symmetric operator A alone.

The batch also proves twelve separated likelihood-ratio classes for two adjacent strict heat preparations by outward interval arithmetic; certifies a bounded two-step segment on all sixty signed continuation branches; gives a rational primal–dual optimum 189/500 for a nonunital negative-orientation qubit channel; and proves a unique continuous minimum-entropy-production selector trajectory inside a declared two-state control model. A concrete nonlinear coordinate change falsifies the ordinary twelfth-response ratio as a natural nonlinear-carrier invariant and identifies a connection or jet splitting as the missing geometric object.

At the refinement level, every self-adjoint exact extension relative to an isometry R is classified as $RAR^* \oplus B$. This turns the previous one-parameter ambiguity into a high-dimensional moduli cone: already a two-child real symmetric refinement has 78 free complement parameters. Repeated spectral-dimension flows consequently depend on unsourced fiber rates and possess a scale orbit. One HMM Chernoff optimizer is only strong numerical evidence, not an interval theorem. No strict source for the localized state, canonical selector or connection, dimensional scale, spacetime, external record, legacy-to-strict completion, Standard Model, gravity, or Theory-of-Everything closure is claimed.

Contents

1	Executive conclusions	1
2	Scope, kernel split, and proof labels	1
3	Program ledger	1
4	ST233–ST235: the second operator and the selector resource	2
4.1	A state-typed noncommuting operator	2
4.2	Twelve interval-separated likelihood-ratio classes	3
4.3	The stabilizer theorem	4
5	ST236: bounded all-branch continuation	5
6	ST237: exact recovery of a nonunital negative-orientation channel	5
7	ST238 and ST244: what selection and reset cost	6
7.1	A verified continuous selector path	6
7.2	Reset moves information; recycling pays	6
8	ST239: a nonlinear-carrier no-go and the missing connection	7
9	ST240–ST243: validation boundaries and statistical inference	7
9.1	Certificate failure is not root loss	7
9.2	Sharp modal minimax constants	7
9.3	Higher-dimensional fixed-factor Clifford projection	8
9.4	Chernoff optimization: intentionally not promoted	8
10	ST245–ST246: complete refinement moduli and spectral dimension	8
10.1	The complete self-adjoint classification	8
10.2	Exact repeated-refinement heat formula	9
11	Falsification ledger	9

12 Relation to FIN ontology and the projection intuition	10
13 Recommended programs ST248–ST262	10
14 Reproducibility	11
15 Final conclusion	11

1 Executive conclusions

The batch closes five questions and sharpens four obstructions.

1. A second noncommuting operator exists naturally *conditional on a state*:

$$\rho \in \Delta_{11} \mapsto M_\rho = \text{diag}(\rho), \quad [A, M_\rho] \neq 0.$$

This is a typed construction, not an arbitrary matrix insertion.

2. The same construction does not solve the selector problem. The state ρ carries the symmetry breaking. A uniform condition retains D_{12} , an even localized condition retains a reflection, and a chiral condition has trivial stabilizer.
3. The ordinary twelfth response is natural under linear carrier changes but not under nonlinear charts. A connection is necessary for a covariant completion.
4. Exact coarse diffusion geometry admits an enormous family of self-adjoint refinements. Geometry preservation alone cannot select a unique deeper layer.
5. The refinement heat trace produces calculable spectral-dimension profiles, but their shape and scale depend on free fiber rates. They do not generate metres, seconds, or a Planck scale.

The deepest surviving interpretation is therefore operational and relational rather than ontologically complete:

$$\boxed{\text{symmetric spectral generator} + \text{typed state} \implies \text{ordered observable layers},}$$

while the provenance of the state and the choice of refinement remain independent data.

2 Scope, kernel split, and proof labels

The strict working object is the frozen twelve-vertex graph Laplacian

$$A = sI - W, \quad W_{xx} = 0, \quad W_{xy} = K_{\text{strict}}(d(x, y)) \quad (x \neq y),$$

with $A\mathbf{1} = 0$ and $A \succeq 0$. The diagonal condition is essential: the quoted row sum s belongs to the off-diagonal weight matrix. The historical ontological kernel $K_{\text{legacy,ont}}$ remains an intermediate bridge kernel. It is not identified silently with K_{strict} , and no legacy physical-role formula is transferred here.

The project ontology treats the nadsoliton as primordial fractal information in a solitonic state, with no separate informational substrate beneath it. This report tests mathematical consequences of the frozen operator only; it does not convert that ontology into a physical theorem.

- **[PROVEN]** denotes an exact deduction or an outward finite certificate within stated premises.
- **[STRONG NUMERICAL EVIDENCE]** denotes reproducible numerical evidence without a complete outward or exact certificate.
- **[CONDITIONAL]** denotes a theorem after introducing the named state, control, channel, connection, bath, or refinement isometry.
- **[BLOCKED]** denotes a required external or ungenerated object.

3 Program ledger

Pro-gram	Object	Result	Boundary
ST233	Typed second operator	[CONDITIONAL] noncommutation and order sensitivity	Localized state is supplied
ST234	Blackwell classes	[PROVEN] 12 disjoint ratio intervals	Two preparations and time supplied
ST235	Conditioned noise	[PROVEN] stabilizer/resource theorem	Does not source the condition
ST236	Branch graph	[PROVEN] 120 local boxes, 30 finite components	Two steps per signed seed only
ST237	Nonunital recovery	[PROVEN] global CPTP optimum 189/500	Pauli-plus-replacer channel
ST238	Continuous selector	[PROVEN] unique optimum and interval cost	Two-state bath/control model supplied
ST239	Nonlinear carrier	[PROVEN] counterexample; [CONDITIONAL] covariant repair	Connection is not derived
ST240	Nuisance subdivision	[PROVEN] 512/512 child boxes	Stratified sample, not full cube
ST241	Correlated minimax	[CONDITIONAL] sharp LAN constants	Asymptotic local experiment
ST242	Clifford polar factor	[PROVEN] unique fixed- X solution	Joint-factor uniqueness open
ST243	HMM Chernoff	[STRONG NUMERICAL EVIDENCE] optimizer near 0.5022593	No outward sum certificate
ST244	Reset ledger	[PROVEN] SWAP bookkeeping and recycling bound	Blank and Hamiltonian supplied
ST245	Refinement moduli	[PROVEN] complete self-adjoint block classification	Local graph constraints open
ST246	Spectral dimension	[PROVEN] heat-trace formula; computed atlases	Fiber rates and scale unsourced
ST247	External evidence	[BLOCKED]	No independently registered events

4 ST233–ST235: the second operator and the selector resource

4.1 A state-typed noncommuting operator

Let ρ be a probability state on the twelve vertices and define the multiplication operator

$$M_\rho f(x) = \rho_x f(x).$$

For $g \in D_{12}$,

$$M_{g\rho} = gM_\rho g^{-1},$$

so the construction is equivariant and correctly typed. Choosing

$$\rho = e^{-0.7A} \delta_0$$

gives a strictly positive state with numerical minimum 0.01662699154 and

$$\|[A, M_\rho]\|_F = 0.2651939971.$$

For comparison, $M_{1/12}$ is scalar and commutes exactly with A .

Proposition 4.1 (Order becomes observable). *For $h > 0$ define*

$$S_{AB}(h) = e^{-hA} e^{-hM_\rho}, \quad S_{BA}(h) = e^{-hM_\rho} e^{-hA}.$$

If $[A, M_\rho] \neq 0$, then the schedules differ. Moreover,

$$\frac{S_{AB}(h) - S_{BA}(h)}{h^2} \longrightarrow [M_\rho, A] \quad (h \downarrow 0).$$

The computed schedule difference decreases quadratically and the scaled value approaches the commutator norm. Thus the state multiplier supplies exactly the kind of second algebra that ST218 proved impossible inside the single- A functional calculus.

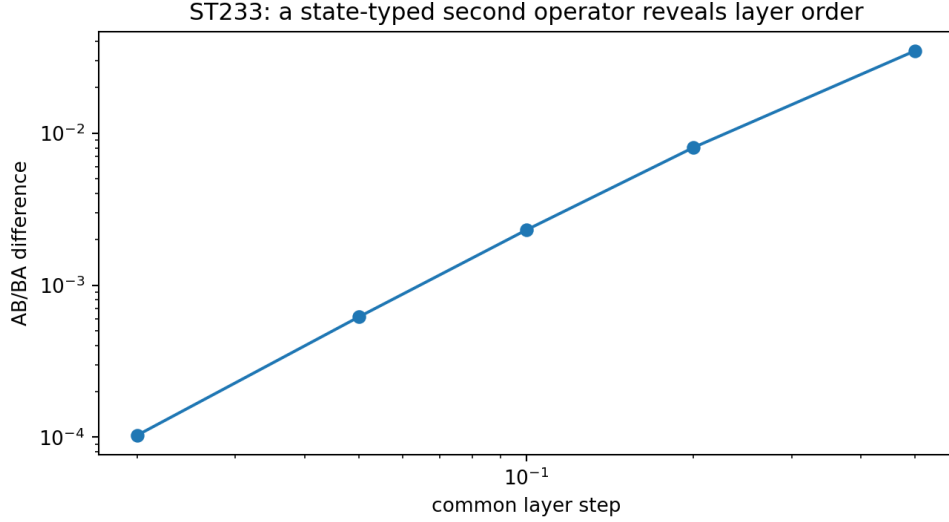


Figure 1: ST233: order sensitivity produced by the state-typed multiplication operator.

4.2 Twelve interval-separated likelihood-ratio classes

ST234 considers the two preparations δ_0 and δ_1 , the vertex measurement, and $e^{-0.7A}$. Each heat entry is evaluated by interval Fourier functional calculus over the complete frozen strict-entry enclosure. All entries remain positive. The twelve likelihood-ratio intervals are pairwise disjoint, with minimum adjacent gap

$$0.02428674898.$$

For a binary experiment, outcomes with identical likelihood ratio are precisely the atoms of the minimal sufficient partition. Hence this declared experiment has twelve atoms throughout the supplied strict enclosure.

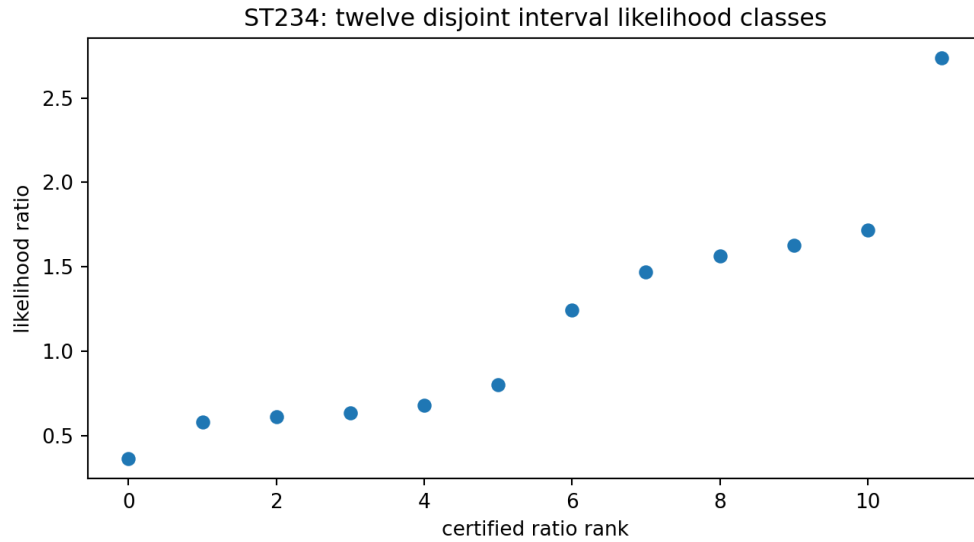


Figure 2: ST234: outward likelihood-ratio intervals, ordered by value. Error bars are much smaller than the plotted symbols.

4.3 The stabilizer theorem

Let an equivariant conditional law satisfy

$$K(g\rho, gE) = K(\rho, E).$$

After fixing ρ , the output can distinguish only orbits of its stabilizer G_ρ . The complete finite calculation gives

ρ	$\mathbf{1}/12$	δ_0	$e^{-0.7A}\delta_0$	chiral localized
$ G_\rho $	24	2	2	1
# vertex orbits	1	7	7	12.

An even localized state selects a vertex but not an orientation. A chiral state supplies both. This is the precise resolution of the apparent paradox: conditioning permits selection, but selection information is in the condition.

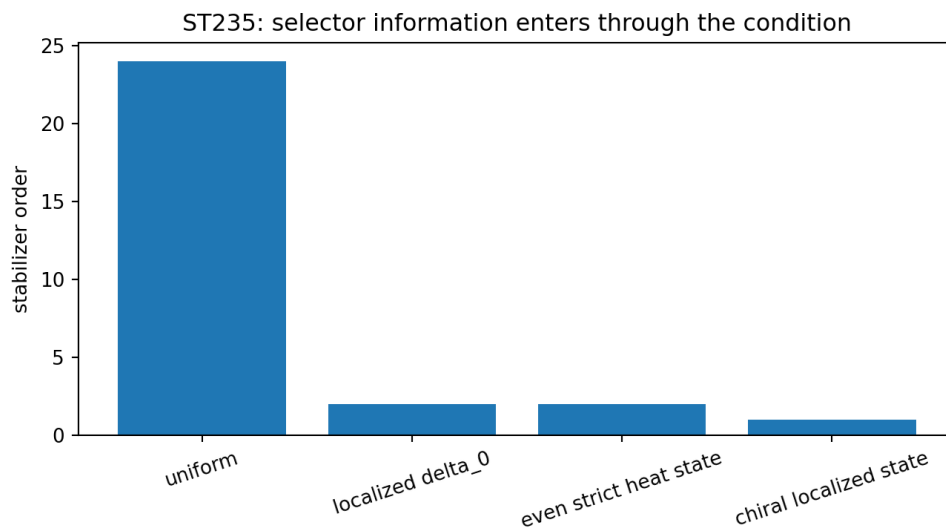


Figure 3: ST235: residual symmetry and distinguishable vertex orbits.

5 ST236: bounded all-branch continuation

The sixty signed seeds inherited from the exact fold campaign were advanced by two pseudo-arclength steps each. Every one of the 120 stationary points has an outward Krawczyk inclusion. The minimum seven-row rank margin is 2.61998×10^{-3} . Opposite signed seeds with the same amplitude and Fourier mode meet at one of thirty exact uniform folds. All boxes from different fold pairs are disjoint, with minimum coordinate clearance 0.00606308.

Theorem 5.1 (Finite certified incidence graph). *The declared set of 120 local boxes and thirty exact fold links has thirty connected components. This statement is exact for the finite graph.*

It is not a theorem about untraced branch segments. A collision, reconnection, or secondary bifurcation outside the two-step windows remains possible.

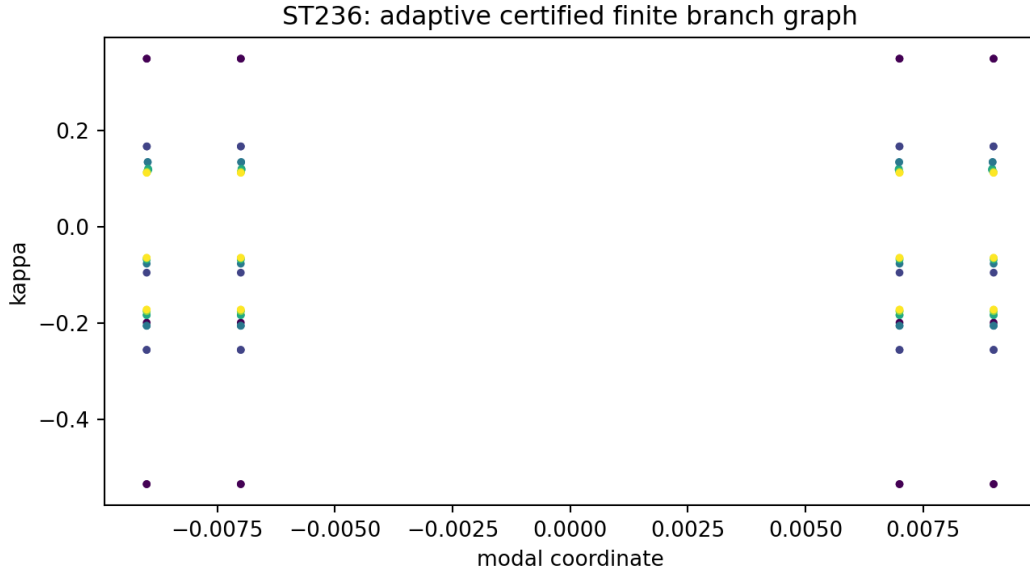


Figure 4: ST236: bounded certified continuation issued from all signed fold seeds.

6 ST237: exact recovery of a nonunital negative-orientation channel

Consider

$$\mathcal{N} = \frac{4}{5}\mathcal{P} + \frac{1}{5}\mathcal{R}_0, \quad \mathcal{R}_0(\rho) = |0\rangle\langle 0| \operatorname{tr} \rho,$$

where the Pauli probabilities are

$$(p_I, p_X, p_Y, p_Z) = \left(\frac{2}{25}, \frac{41}{100}, \frac{29}{100}, \frac{11}{50} \right).$$

Its Bloch linear part has determinant -0.00106496 and translation $(0, 0, 1/5)$, so the example is both negative-orientation and genuinely nonunital.

Theorem 6.1 (Global CPTP recovery optimum). *For every trace-preserving recovery $\mathcal{R}$,*

$$F_e(\mathcal{R} \circ \mathcal{N}) \leq \frac{4}{5} \frac{41}{100} + \frac{1}{5} \frac{1}{4} = \frac{189}{500}.$$

The unitary X recovery attains the bound. A rational Choi primal and a dual matrix have equal value; the computed dual slack eigenvalues are

$$(0, 0.048, 0.076, 0.132)$$

up to 2.1×10^{-17} floating replay error.

The theorem is global over all CPTP recoveries for this channel family, not for arbitrary affine qubit channels.

7 ST238 and ST244: what selection and reset cost

7.1 A verified continuous selector path

The declared two-state reduction obeys

$$\dot{x} = \pi - x, \quad 0 \leq \pi \leq 1,$$

and uses the Markov entropy-production density

$$L(x, \pi) = (\pi - x)(\text{logit } \pi - \text{logit } x).$$

For $x(0) = 1/12$, $x(4) = 0.6$, the unique continuous optimum has first integral

$$C = \frac{\dot{x}^2}{\pi(1-\pi)}, \quad \pi(x, C) = \frac{2x + C + \sqrt{C^2 + 4Cx(1-x)}}{2(1+C)}.$$

Outward quadrature proves

$$C \in [0.0744, 0.0747], \quad \Sigma \in [0.3176910887, 0.3191979592].$$

The travel-time interval at $C = 0.0744$ lies strictly above four, while the interval at $C = 0.0747$ lies strictly below four.

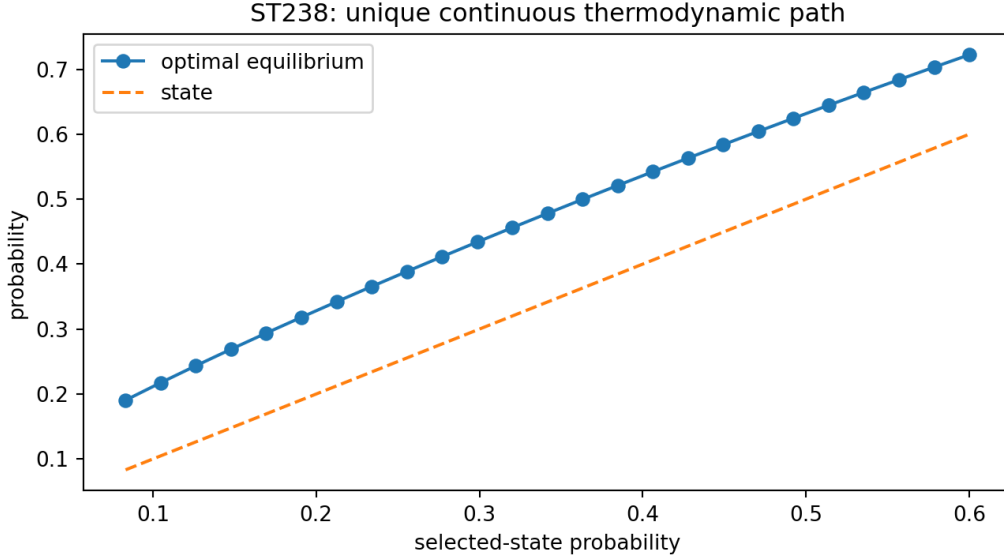


Figure 5: ST238: optimal equilibrium control and the induced monotone state trajectory.

This prices a selector trajectory only after supplying the bath, preferred vertex, two-state reduction, maximum rate, endpoints, and dimensionless time. It does not generate the selector or convert the cost into joules.

7.2 Reset moves information; recycling pays

For identical register Hamiltonians, the global SWAP implements

$$\rho \otimes |0\rangle\langle 0| \mapsto |0\rangle\langle 0| \otimes \rho.$$

The joint spectrum, entropy, and total energy are unchanged. Hence the reversible logical step has $\Delta F_{\text{total}} = 0$: information is transferred to the ancilla, not erased. Restoring the blank requires at least

$$W \geq F_{\beta}(|0\rangle\langle 0|) - F_{\beta}(\rho).$$

For a degenerate Hamiltonian and $\beta = 1$, this is $S(\rho)$ and reaches $\log 16$ for the maximally mixed input. The controller work and finite-time friction remain unpriced.

8 ST239: a nonlinear-carrier no-go and the missing connection

Take

$$V(x) = \frac{a}{2}x^2 + \frac{g}{12}x^{12}, \quad x = y + cy^6.$$

At the origin the Hessian and tangent scale are unchanged, yet

$$\frac{D_y^{12}V}{11!} = g + 6ac^2.$$

For $(a, g, c) = (1.7, 0.4, 0.3)$, the ordinary normalized response changes from 0.01657168 to 0.05460368. Thus ordinary higher derivatives do not define a nonlinear carrier-natural tensor.

Theorem 8.1 (Necessary geometric completion). *To transport the twelfth response naturally under nonlinear carrier charts, one must add a connection or an equivalent jet splitting. Conditional on a transported connection,*

$$\mathcal{I}_{12}^\nabla = \frac{(\nabla^{12}V)(v^{\otimes 12})}{11! [(\nabla^2 V)(v, v)]^6}$$

is a scalar under chart change.

The connection is a newly identified missing object; no canonical strict source for it is known.

9 ST240–ST243: validation boundaries and statistical inference

9.1 Certificate failure is not root loss

At global nuisance halfwidth 5×10^{-4} , 64 deterministic cells selected from four strata per coordinate all fail the coarse Krawczyk test with local halfwidth 2.0833×10^{-5} . Splitting each into eight children reduces the local halfwidth to 1.0417×10^{-5} ; all 512 children pass, with minimum margin 0.001779.

Therefore these are certificate-resolution failures, not root-loss events in those boxes. The 64 parents do not cover the full cube, and the result must not be reported as a full 5×10^{-4} continuation theorem.

9.2 Sharp modal minimax constants

Within the declared regular six-parameter local experiment, local asymptotic normality gives

$$R_{\text{trace}} \sim \frac{\text{tr}(I^{-1})}{n}, \quad R_{\text{worst mode}} \sim \frac{\max_k I_k^{-1}}{n}.$$

For exchangeable clusters of size 20 and intracluster correlation 0.05, both constants are multiplied by

$$1 + (20 - 1)0.05 = 1.95.$$

This is an asymptotic conditional theorem, not a finite-sample detector guarantee.

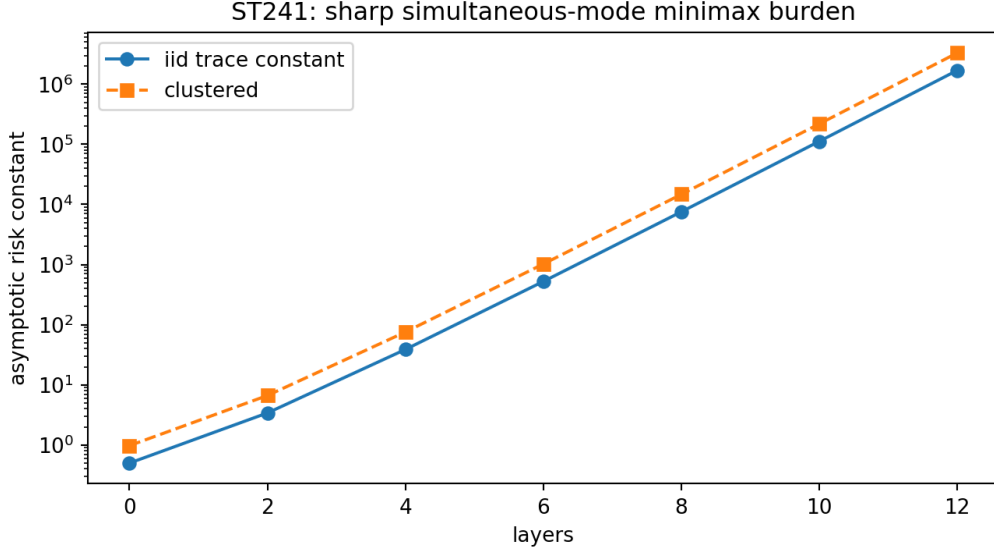


Figure 6: ST241: trace-risk constants and their cluster inflation across developed layers.

9.3 Higher-dimensional fixed-factor Clifford projection

Fix the balanced involution $X = \text{diag}(I_n, -I_n)$. Every Hermitian involution anticommuting with X has form

$$Z(U) = \begin{pmatrix} 0 & U \\ U^* & 0 \end{pmatrix}, \quad U \in U(n).$$

Projecting a target first to its X -odd block C reduces the nearest-factor problem to unitary Procrustes. If C is invertible, its polar unitary $U = C(C^*C)^{-1/2}$ is the unique global minimizer. Numerical replays for $n = 2, 3, 4$ have unitarity and involution errors below 2.1×10^{-15} . This theorem fixes X first; simultaneous global uniqueness of both factors is not implied.

9.4 Chernoff optimization: intentionally not promoted

For all $2^{16} = 65,536$ records of the supplied binary HMM, long-double summation gives

$$s_* \approx 0.50225930327, \quad Z_{16}(s_*) \approx 6.69672763505 \times 10^{-6}.$$

The derivative changes sign across $[0.5022593, 0.5022594]$, and the optimized coefficient is smaller than the $s = 1/2$ coefficient $6.69881023894 \times 10^{-6}$. Exact strict convexity of Z_{16} proves that a true derivative zero is unique. However, the endpoint sums were not enclosed by outward interval arithmetic. The location and the induced asymptotic candidate bracket $[0.68990, 0.81987]$ are therefore **[STRONG NUMERICAL EVIDENCE]**, not a proof certificate.

10 ST245–ST246: complete refinement moduli and spectral dimension

10.1 The complete self-adjoint classification

Let $R : \mathbb{R}^{12} \rightarrow \mathbb{R}^{12q}$ be an isometry and suppose

$$\tilde{A}R = RA, \quad \tilde{A} = \tilde{A}^*.$$

Self-adjointness makes $\text{Ran } R$ reducing. Hence

$$\boxed{\tilde{A} = RAR^* \oplus B \quad \text{on } \text{Ran } R \oplus (\text{Ran } R)^\perp.}$$

Conversely every self-adjoint B yields exact functional-calculus intertwining, and $B \succeq 0$ preserves positivity. With $m = 12(q - 1)$, the real symmetric moduli dimension is $m(m + 1)/2$:

q	2	3	4	5
m	12	24	36	48
$\dim \text{Sym}(m)$	78	300	666	1176.

Three independent $q = 3$ replays preserve generator intertwining and coarse heat distances to errors below 10^{-14} . The earlier one-rate Kronecker refinement is only a tiny slice of this cone.

10.2 Exact repeated-refinement heat formula

For two-state fiber gaps μ_j , heat traces multiply under Kronecker sums. Therefore

$$d_s(t) = 2t\langle\lambda\rangle_t + \sum_j \frac{4\mu_j t}{e^{2\mu_j t} + 1}.$$

Constant rates create a growing intermediate peak; geometrically separated rates create staged contributions. Neither is selected by coarse FIN geometry. Moreover,

$$\mu_j \mapsto c\mu_j, qquad t \mapsto t/c$$

leaves every fiber contribution invariant. Frequency and spectral flow supply dimensionless phase/time ratios only; they do not select seconds or lengths.

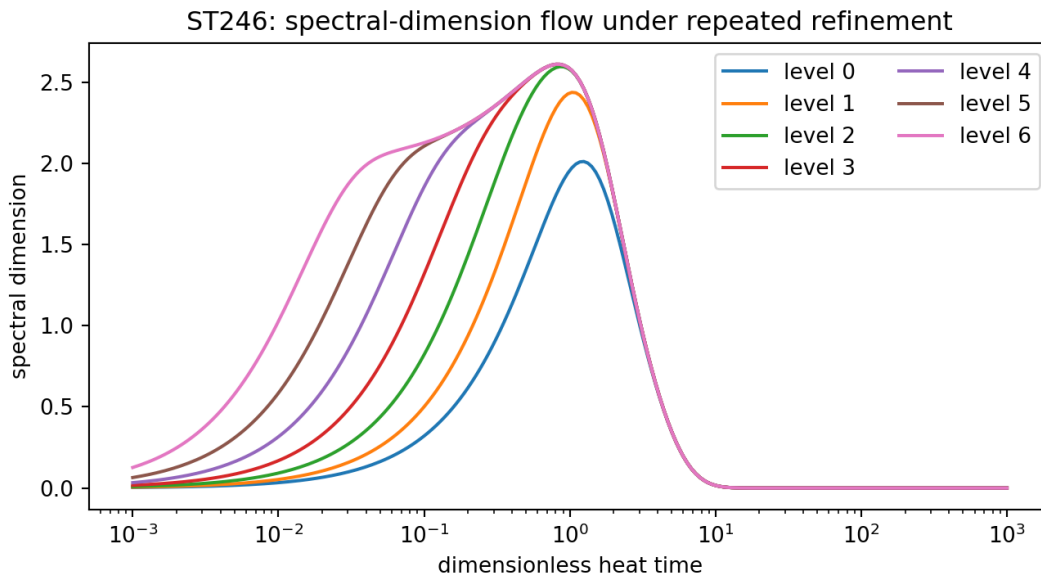


Figure 7: ST246: spectral-dimension flows for constant and geometric fiber-rate choices.

11 Falsification ledger

Candidate claim	Falsification or surviving result	Epistemic consequence
One symmetric operator contains layer order	Every $f(A)$ commutes; ST233 needs M_ρ	Second typed object is necessary
Conditioned noise generates a selector	Stabilizer information is inherited from ρ	State provenance remains open

Linear response naturality extends nonlinearly	Explicit $x = y + cy^6$ counterexample	Connection/jet splitting required
Coarse certificate failure signals root loss	512 child boxes all pass	Numerical method failure separated from ontology
HMM optimizer is interval certified	Endpoint sums are long-double, not outward	Demoted to strong numerical evidence
Exact diffusion refinement is essentially unique	Full $RAR^* \oplus B$ classification	Large nonuniqueness moduli proved
Repeated refinement generates physical scale	Fiber rates free; simultaneous scale orbit	No seconds, lengths, or Planck scale
Local computation can create external evidence	No registered laboratory record exists	ST247 remains blocked

12 Relation to FIN ontology and the projection intuition

The results are compatible with an information-first interpretation only in a limited, mathematical sense. A single spectral generator supports several functional channels, and a state-typed multiplication algebra makes order and preparation operationally visible. Refinement can hide arbitrary complement dynamics from all coarse observables. These facts make a “deeper state projected to a coarser observation” architecture mathematically possible.

They do not prove that the observed universe is such a projection. In fact, ST245 makes the inverse problem harder: infinitely many deeper operators yield exactly the same coarse heat geometry. To turn layered fractal compression into physics, one needs a rule that selects R , B , a state, an instrument, a clock, and dimensional calibration, followed by observables that distinguish the selected refinement from alternatives. Seeing lower scales through fractal layers is therefore a research hypothesis, not a consequence of the current strict kernel.

Likewise, symmetry is a reservoir of equivalent possibilities, not positive feedback by itself. Positive feedback requires a dynamical gain or instability. A state with reduced stabilizer can select and amplify one branch, but ST235 proves that the asymmetry is then carried by that state unless a strict internal source is found. QW-2191 remains open.

13 Recommended programs ST248–ST262

- ST248.** Search for a strict source of the preparation entering M_ρ , or prove a preparation-source no-go.
- ST249.** Classify all D_{12} -natural typed second operators in the vertex multiplication and commutant algebras, with their minimal conditioning data.
- ST250.** Extend the branch graph only until an interval-certified collision, closure, or declared global resource stop.
- ST251.** Solve one generic rational affine-qubit recovery instance outside the Pauli-plus-replacer symmetry.
- ST252.** Add endpoint preparation and finite-rate actuator costs to the continuous selector optimum.
- ST253.** Search for a canonical connection from strict response data and apply the ST239 obstruction test.
- ST254.** Complete the halfwidth-0.0005 nuisance shell by adaptive covering, or stop at a predeclared resource boundary.
- ST255.** Construct finite-sample estimators approaching the ST241 correlated minimax constants.
- ST256.** Prove or refute global joint higher-dimensional Clifford-factor uniqueness under two simultaneous spectral gaps.
- ST257.** Replace the block Chernoff comparison by a certified belief-transfer spectral-radius enclosure.

- ST258.** Include pulse-controller work and finite-temperature blank preparation in reset bookkeeping.
- ST259.** Intersect the ST245 moduli cone with graph-Laplacian positivity, locality, and declared symmetry constraints.
- ST260.** Test whether any internally selected fiber-rate distribution yields a stable spectral-dimension plateau.
- ST261.** Derive operational observables capable of distinguishing two members of the refinement moduli cone.
- ST262.** Resume external validation only after independently registered laboratory events, calibration hashes, and custody records exist.

The top priority is ST248. Without a source for the state, the newly constructed second operator is an operational completion, not a strict-core derivation. ST253 and ST259 are the next most informative obstruction tests: they ask whether the new connection and the large refinement cone can be reduced by already available strict data.

14 Reproducibility

The executable source is `fin_st233_st247_research.py`. It writes one aggregate JSON file, one CSV ledger, fifteen per-program JSON packets, and seven figures. The lightweight independent replay `test_fin_st233_st247.py` checks packet hashes, live noncommutation, exact rational identities, interval ordering, all stored continuation inclusions, status discipline, and the epistemic boundary. The SHA-256 manifest fixes the release byte-for-byte.

The numerical environment recorded in the aggregate JSON includes Python, NumPy, SciPy, and SymPy versions. ST234, ST236, ST238, and ST240 use outward interval/Krawczyk logic; ST243 explicitly does not and is labeled accordingly. Random demonstrations use the fixed seed 20260812.

15 Final conclusion

The new research does not close FIN as physics. It does identify two missing theoretical objects more precisely than before:

$$\left[\begin{array}{l} \text{a sourced state/preparation for the second algebra} \\ \text{a canonical connection for nonlinear carrier response} \end{array} \right].$$

It also proves that a third hoped-for shortcut fails: exact preservation of coarse diffusion geometry cannot choose a unique fractal refinement because the complement generator B remains free in a high-dimensional cone.

The strongest defensible present statement is this: FIN’s strict operator admits a coherent operational extension in which state, layer order, measurement discrimination, control cost, and multiscale refinement can be treated rigorously. The same mathematics proves that the state, nonlinear connection, dimensional scale, and deeper refinement are not yet inevitable consequences of the strict operator. That boundary, rather than a claim of physical closure, is the scientific result of ST233–ST247.